higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

## MATHEMATICAL LITERACY
(First Paper)
**NQF LEVEL 3**

(10401023)

**20 February 2019 (X-Paper)**
**09:00–12:00**

**Calculators may be used.**

**This question paper consists of 9 pages and 2 answer sheets.**

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**TIME: 3 HOURS**
**MARKS: 150**

## INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. You may use an approved calculator (nonprogrammable and nongraphic), unless stated otherwise.
5. Show ALL calculations clearly.
6. Round off your answers correctly according to the given context. In all other cases, where the context is not specific, round off your answers correctly to two decimal places.
7. Indicate units of measurement, where applicable.
8. Maps and diagrams are NOT necessarily drawn to scale, unless stated otherwise.
9. Answer QUESTIONS 4.2.3 and 5.5 and 5.6 on the attached ANSWER SHEETS. Write your EXAMINATION NUMBER in the space provided and hand in the ANSWER SHEET with the ANSWER BOOK.
10. Write neatly and legibly.

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## QUESTION 1

1.1 Calculate the following **without** using a calculator. Show **ALL** working.

1.1.1 $8 + (3{,}6 + 3{,}94) \times 2 \div (4 + 3)$ (4)

1.1.2 $\sqrt{\frac{64}{16}}$ (2)

1.2 Write each of the following as a common fraction in its simplest form:

1.2.1 7 days out of the month of February 2017. February 2017 had 28 days.

1.2.2 36 seconds out of an hour

1.2.3 25 cm out of 1 m

(3 × 2) (6)

1.3 Water is pumped into an empty irrigation reservoir (dam) at a rate of 900 ℓ/h. The capacity of the reservoir is 27 000ℓ.

1.3.1 How long will it take to fill the reservoir? Give your answer in days and hours. (3)

1.3.2 If the pump works for 18 hours, calculate the percentage of the reservoir that will be filled? (4)

1.4 Thabo scored a rating of $\frac{22}{25}$ while Esther scored a rating of $\frac{43}{50}$ for a job interview.

1.4.1 Who scored the better rating? Show your calculations. (2)

1.4.2 Write the ratings of Thabo and Esther as a ratio Thabo : Esther in its simplest form. (2)

1.5 Two employees had to complete a task. Employee A completed the task in 1 hour 23 minutes and 12 seconds while employee B took 1 hour 39 minutes and 4 seconds to complete the task. Calculate the difference in the time taken by the employees to complete the task. (3)

1.6 The oven thermometer on the right measures temperature in °C.

Write down the reading of the oven thermometer as displayed, in °C.




(2)

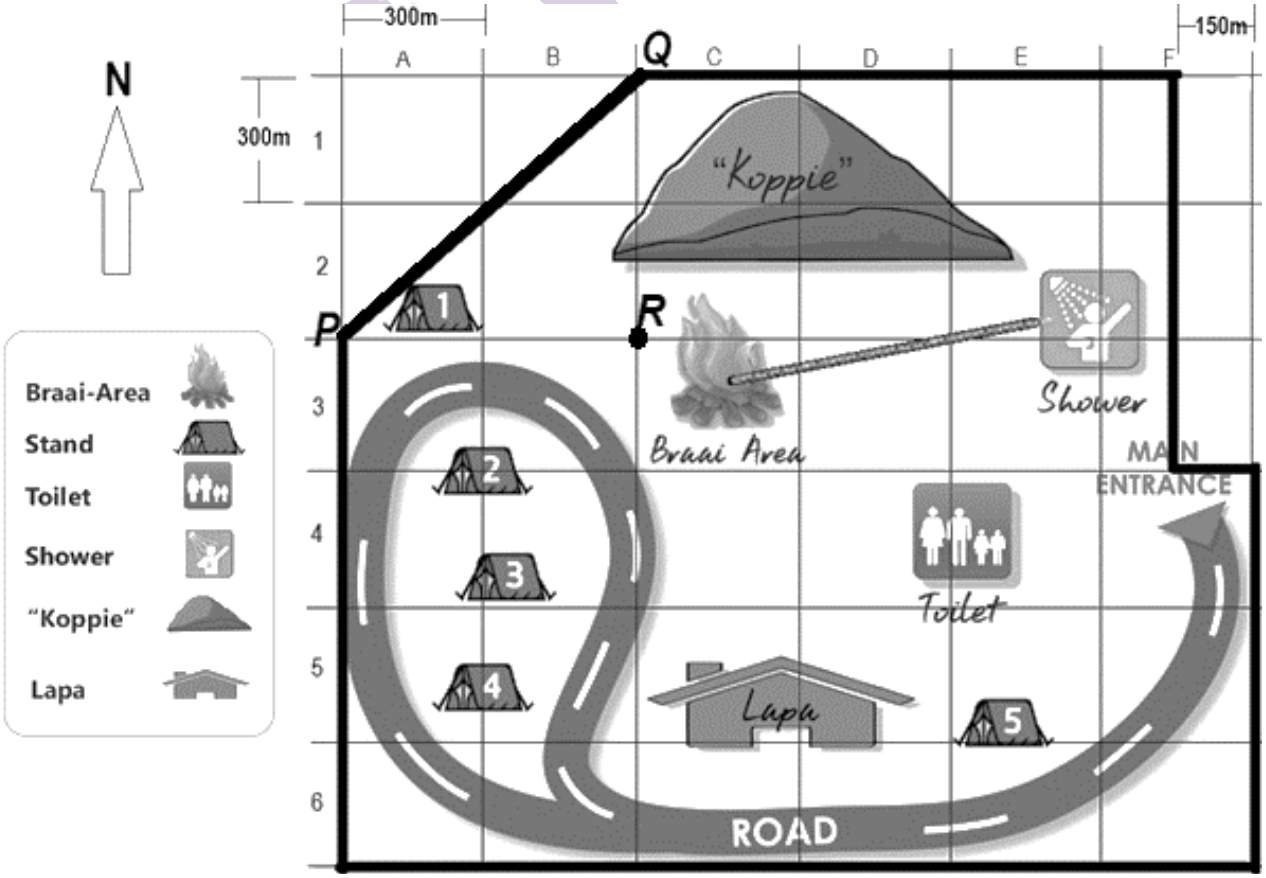
1.7 Convert 200 °C into °F.

Formula: $°F = °C \times \frac{9}{5} + 32$

(2)
**[30]**

**QUESTION 2**

2.1 The grid map below, shows the layout and the directions of a camping site on a game reserve. The map shows the position of the following facilities: braai area, stands (for rest camps), toilets, shower, 'koppie' and Lapa. The main entrance and the road on which the visitors may drive also appear on the map. Study the grid map and answer the questions that follow.



2.1.1 Name the facilities that are found in cell E2. (2)

2.1.2 What is the grid reference of Stand 4? (1)

2.1.3 What is the compass direction of the Lapa from Stand 1? (1)

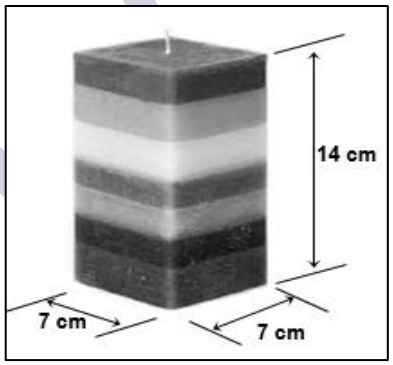
2.1.4 Use Pythagoras' theorem and rounding off to the nearest metre, to prove that the length of PQ is equal to 849 m.

Formula: $PQ^2 = QR^2 + PR^2$ (4)

2.1.5 The camping site is fenced with game mesh fencing. The fence and the entrance gate are indicated on the map by a black line. The entrance gate is made of square tubing and it is 10 metres wide. Determine the length of the game mesh fencing, in kilometres.

 (8)

2.2 Lucy produces candles for decorations at functions. Given alongside is a picture of a candle. The candle is made up of seven equal differently coloured portions of wax.




The candle has the length and breadth of 7 cm each and a height of 14 cm.

Study the picture of the candle and answer the following questions.

2.2.1 What is the shape of the base of the candle? (1)

2.2.2 Calculate the area of the base of the candle.

Formula: Area = Length × Breadth or Area = $L^2$. (2)

2.2.3 Determine the amount of wax required to make the entire candle in $cm^3$.

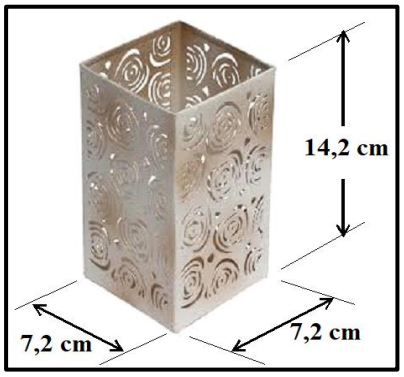
Formula: Volume = area of the base × height. (3)

2.2.4 Determine the amount of wax required to make one out of the seven coloured parts of the candle. (2)

2.2.5 Lucy also designs candle holders which are used to cover the sides and the bottom of the candle. The cost of the material she uses is 5 cents per $cm^2$.

Calculate the cost in rand of the material needed to produce a candle holder.

Formula:
Cost = Area × Charge/$cm^2$




(6)
**[30]**

## QUESTION 3

3.1 Most mobile phone operators in South Africa offer a 'per second' tariff plan to their prepaid customers. The table below shows an example of how much a prepaid customer would have to pay based on the tariff plan chosen and the duration of a call.

| **Tariff Plan Name** | **Any Time Per Second** | **Pay Per Second** | **66c on Prepaid** | **Sim–Sonke** |
|---|---|---|---|---|
| **Per Minute Fee** | **R1,20** | **R0,79** | **R0,66** | **R0,75** |
| Length of a call (minutes) | Total cost | Total cost | Total cost | Total cost |
| 0,5 | R0,60 | R0,40 | R0,33 | **B** |
| 1 | R1,20 | R0,79 | R0,66 | R0,75 |
| 1,5 | **A** | R1,19 | R0,99 | R1,13 |
| | | | | |

*Source:[https://www.icasa.org.za/uploads/files/Bi-Annual-Retail-Tariffs-Report-Q4-2017.pdf]*

Use the above tariff table to answer the questions that follow:

3.1.1 Calculate the actual cost per second of each of the following tariff plans:

Give your answer in cents.

(a) Any Time Per Second (2)

(b) Pay Per Second (2)

3.1.2 Calculate the value of A, the total cost, in rand, of a 1,5-minute call on 'Any Time Per Second' tariff plan. (3)

3.1.3 Calculate the value of B, the total cost, in rand, of a 0,5-minute call on 'Sim-Sonke' tariff plan. (3)

3.1.4 Calculate the total call time, in minutes, for an amount of R12,00 on an 'Any Time Per Second' tariff plan. (2)

3.1.5 Calculate the total cost, in rand, for a 50 second call on a '66c on Prepaid' tariff plan. (2)

3.1.6 Calculate the difference in cost between a 'Pay Per Second' tariff plan and a 'Sim–Sonke' tariff plan for a call that lasts for 120 seconds. (5)

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3.2 Sandy decides to open a current bank account for her small hairdressing business. She inspects the following bank charges, of two banks, for a small business account:

| TRANSACTION | Fees/Charges: ABS BANK | Fees/Charges: BUDGET BANK |
|---|---|---|
| Deposits: Cash ATM | R 4,20 plus R 1,25 per R 100 or part thereof | R 10 |
| Withdrawals: Own ATM | R 3,50 plus R 1,25 per R 100 | R 7 |
| Debit orders/stop orders | R 5,10 for first R 100, thereafter an additional R 2,50 per R 100 or part thereof, maximum of R 50 | R 5 for first R 100, thereafter an additional R 2 per R 100 or part thereof, max of R 34 |
| Electronic transfers | R 2 for first R 100, thereafter an additional R 1,50 per R 100 or part thereof, max of R 20 | R 2 for first R 100, thereafter an additional R 1 per R 100, or part thereof, max of R 12 |
| Enquiries: statements of balance | First free, thereafter R5 per enquiry | R 2,50 per enquiry |

Study the above bank charges to answer the questions that follow:

3.2.1 How much will Budget Bank charge for a cash deposit of R1 525 at one of its ATMs? (1)

3.2.2 Calculate the amount that ABS Bank will charge for a cash deposit of R1 525 at one of its ATMs. (3)

3.2.3 Which bank charges a cheaper debit order fee for an amount of R 1 500 from one of its ATMs? Show all calculations? (4)

3.2.4 Sandy plans to make 4 bank statement enquiries per month. Which bank charges a cheaper enquiry fee? Show all calculations? (3)

**[30]**

**QUESTION 4**

Sandy hires a young hairstylist and offers the hairstylist the following:

- a basic wage of R1 000 per month
- an additional R50 commission for every customer's hair she styles during that month
- the hairstylist must work for 20 days a month




4.1 Calculate the following:

4.1.1 Calculate the hairstylist's basic wage per day. (2)

4.1.2 If the hairstylist styled only 4 customers hair on a day, how much would she earn for that day? (2)

4.1.3 The hairstylist earned R700 on a busy Saturday. How many customers did the hairstylist style on that day? (3)

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4.2 The table below shows the total monthly wage of the hairstylist based on the number of customers hair she styles. Assume that the hairstylist works a complete 20-day month.

| Number of customers | 0 | 20 | 40 | 60 | 120 |
|---|---|---|---|---|---|
| Total wage per month in rand | R1 000 | R2 000 | R3 000 | R4 000 | **A** |

Study the above table to answer the questions that follow:

4.2.1 Determine the value of A. (2)

4.2.2 Derive a formula to determine the total wage per month. (3)

4.2.3 Use the above table on the hairstylist's total wage per month to draw and label a line graph on the grid found on ANSWER SHEET 1 (attached). Provide a suitable heading for the graph. (7)

4.2.4 Use the graph and the table in QUESTION 4.2.3 to choose the correct answer from those given in brackets. Write only the word or words next to the question number (4.2.4a–4.2.4c) in the ANSWER BOOK.

(a) The independent variable is the (wages per month/number of customers).

(b) This graph is an example of a (direct/inverse) proportion.

(c) This is an (increasing/decreasing) graph

(3 × 1) (3)

4.3 The hairstylist bought a multi journey monthly bus fare ticket for R356,40. The ticket allows her a maximum of 44 trips per month. The relative bus fare per trip depends on the total number of bus trips that the hairstylist will make during that month.

4.3.1 Determine the relative bus fare per trip if the hairstylist made 40 bus trips in that month. (3)

4.3.2 Explain why the relationship between the number of bus trips taken and the relative bus fare per trip is inversely proportional. (2)

4.3.3 The cash bus fare for people who do not buy multi journey bus tickets is R10,80 per single trip. If the hairstylist made 44 trips in that month, determine how much she saved by purchasing the multi journey ticket.

FORMULA: Total cash bus fare for single trips = R10,80 × number of single trips (3)
**[30]**

## QUESTION 5

Precious has a small catering business. When she has functions, she hires 16 students as casual workers. They are paid according to the number of hours that they work and the kind of task they perform. Below are the wages her workers earned for December 2017 and January 2018:

| December 2017 wages | | | | | | | |
|---|---|---|---|---|---|---|---|
| R496 | R626 | R399 | R496 | R615 | R591 | R650 | R615 |
| R710 | R830 | R615 | R900 | R750 | R584 | R564 | R699 |

| January 2018 wages | | | | | | | |
|---|---|---|---|---|---|---|---|
| R425 | R601 | R450 | R394 | R455 | R450 | R550 | R399 |
| R490 | R385 | R605 | R602 | R641 | R421 | R650 | R374 |

Study the above table to answer the questions that follow:

5.1 What is the mode of wages paid in December 2017? (2)

5.2 Calculate the range of the wages paid in December 2017. (2)

5.3 The mean (average) of the wages paid in December 2017 was R633,75. Answer the following questions:

5.3.1 Show how the mean of R633,75 was calculated. (2)

5.3.2 How many casual workers earned more than the mean wage for December 2017? (2)

5.3.3 Calculate the percentage of workers who earned less than the mean wage for December 2017. (3)

5.3.4 Calculate the difference between the mean wages paid for December 2017 and January 2018. (4)

5.4 Determine the median of the wages paid in January 2018. (3)

5.5 Use the information from the tables above to complete the frequency table, for wages earned in December 2017 and January 2018, found on ANSWER SHEET 1 (attached). (6)

5.6 Use the frequency table to draw a double bar graph on the grid found on ANSWER SHEET 2 (attached). Provide a suitable legend for the graph. (6)

**[30]**

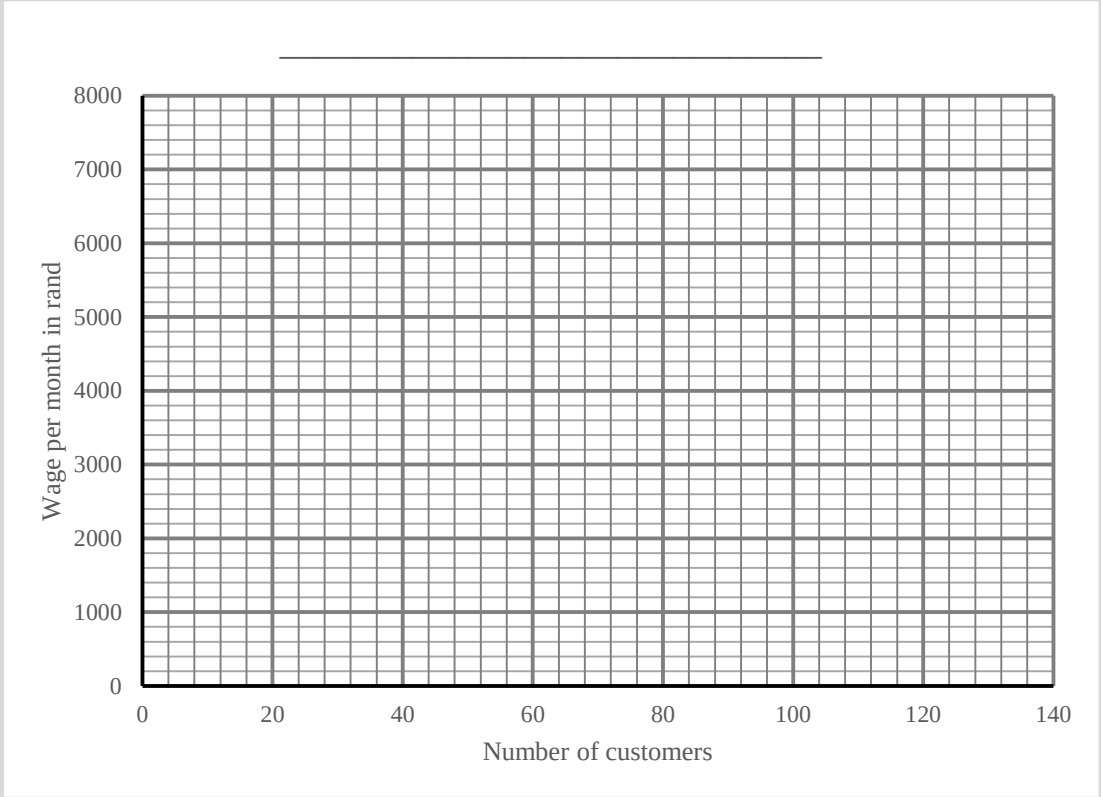
**TOTAL: 150**

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**ANSWER SHEET 1** **EXAMINATION NUMBER:**

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | |

**QUESTION 4.2.3**




(7)

**QUESTION 5.5**

| **Wages earned in December 2017 and January 2018** | | |
|---|---|---|
| **ITEM (in rand)** | **FREQUENCY (December 2017)** | **FREQUENCY (January 2018)** |
| 300 – 399 | | |
| 400 – 499 | | |
| 500 – 599 | **3** | **1** |
| 600 – 699 | | |
| 700 – 799 | 2 | 0 |
| 800 – 899 | 1 | 0 |
| 900 – 999 | 1 | 0 |

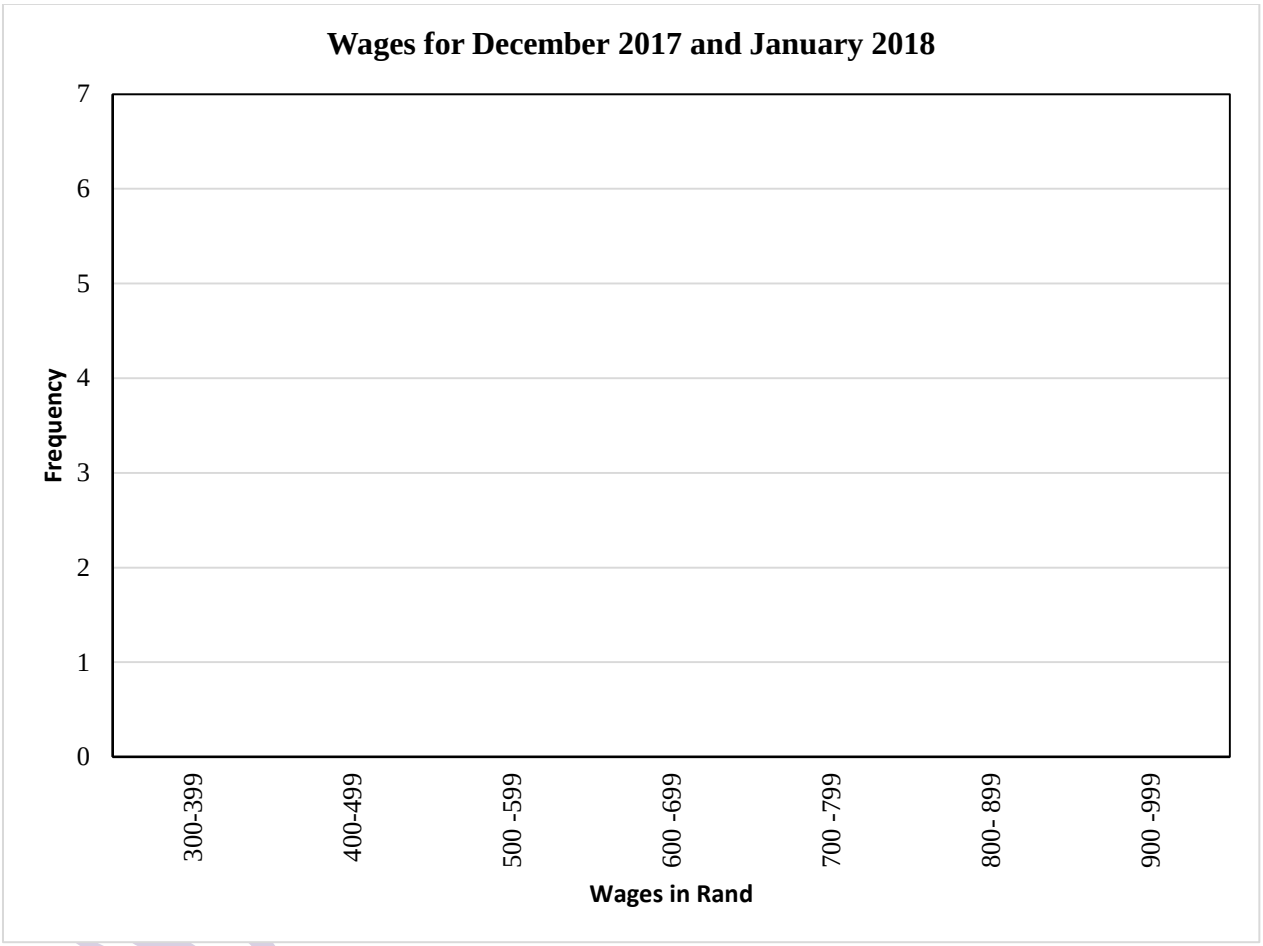
(6)

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**ANSWER SHEET 2** **EXAMINATION NUMBER:**

**QUESTION 5.6**




**Legend:**

(6)

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